

Agenda:

① How to perform text preprocessing

- HTML Tags remove
- punctuation remove
- emoji Handle
- Lemmatization
- Stemming
- lower case

② Text Representation / word Embeddings / Text vectorization

- One Hot Encoding
- Bow (Bag of word)
- TFIDF
- word2vec

③ project → Sentiment Analysis using ML Approach

- ML Algo
-

* Lower case

① My name is Bappy

② bappy is a Data scientist

↳ Bappy ↳ bappy
↳ Bappy ↳ bappy →

You chat on WhatsApp, Instagram, Telegram

* I want the money as soon as possible

+ I want the money ASAP →

Clutch Amazon customer → clutch
→ raising question

This pizza is Amzing → Amazing
spell mistake

dependent
This pizza is very testy I am very happy
To be here in this store, Awesome experience
positive

Increase ↓ in → computation

This movie is Amazing 😊
positive

same
playing, played, plays

starty → Play

😊 😊 ❤️ → positive

ID

Text Representation

corpus

vocabulary

Document

Words

→ R_1

→ R_2

→ R_3

→ R_4

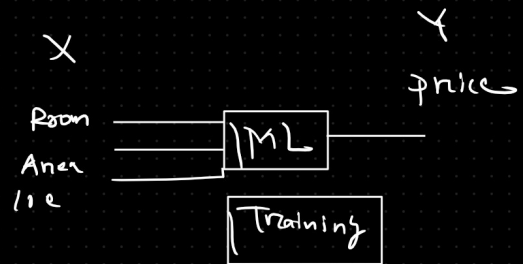
→ R_5

Natural language processing is an interdisciplinary subfield of computer science and linguistics. It is primarily concerned with giving computers the ability to support and manipulate speech.

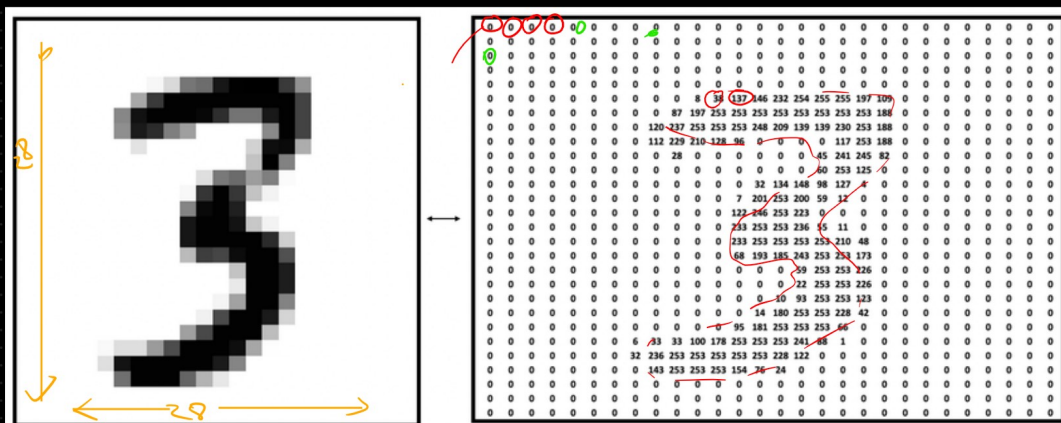
- ① What is feature Extraction from text
- ② Why we need it?
- ③ Why it is so difficult
- ④ What is the core idea?
- ⑤ What are the techniques there?

ML: tabular data, csv, xsl

# Room	Area	Location	Price
5	20	Delhi	5cr
-	-	-	-
-	-	-	-
-	-	-	-

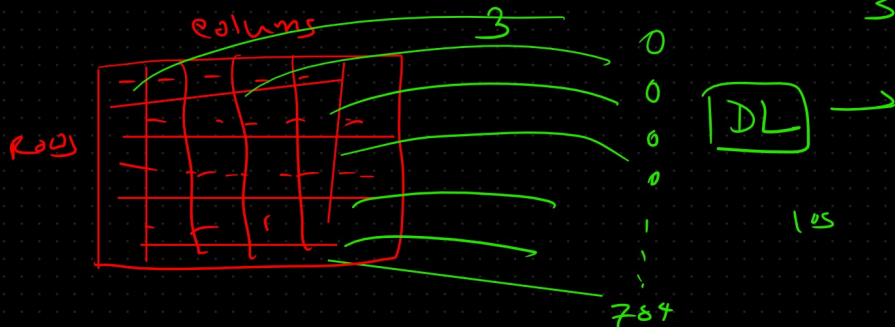


CV: Images, Videos,



28x28

28x28
= 784



One Hot:

D ₁	people watch dswithbary
D ₂	dswithbary water dswithbary
D ₃	people write comments
D ₄	dswithbary write comment

corpus:

people watch dswithbary dswithbary
water dswithbary people write
comments dswithbary write comment

to
20
30

D₁ = $\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \end{bmatrix}$

3x5

0x0
0
0 0 0

N = 5

people	water	dswithbary	write	comment
1	0	0	0	0
0	1	0	0	0
0	0	1	0	0
0	0	1	0	0
0	0	1	0	0
0	0	1	0	0
0	0	1	0	0
0	0	1	0	0
0	0	1	0	0
0	0	1	0	0

D₂ = $\begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \end{bmatrix}$

Sent 1 → $\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \end{bmatrix}$

3x5 $\begin{bmatrix} 4 \\ 3 \end{bmatrix}$ model → shape

My name is Bappy — $\begin{bmatrix} 0 & 0 & 0 & 1 \end{bmatrix}$

* Drawbacks:

- ① Sparsity
- ② No fixed size
- ③ OOV
- ④ No capturing semantic meaning

* POS: easy to implement

Bag of words

		X	Y
1	D ₁	people watch dsalibang	1
2	D ₂	dsalibang watch dsalibang	1
3	D ₃	people write comments	0
4	D ₄	dsalibang write comment	0

Researcher prove

during sentiment Analysis

Bag of word is best technique

people	watch	dsalibang	write	comment
1	1	1	0	0
0	1	2	0	0
1	1	0	1	1
0	0	1	1	1

This movie is Wow - ¹
positive → positive

This movie is Amazing. Amazing performance by SRK. Wow!

Very bad movie. bad performance by X --
Negative

This movie is Amazing. Amazing perform by SRK
But very bad performance by Deepika →

1

n-grams → 3

This movie is very good → positive

This movie is not very good → Negative